

CODEMIX SKILL

The safety + decision layer between AI agents and their actions

*“Order cancel **mat** karo, bas address change karna hai.”*



We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

1

Initiative Overview

Initiative: Codemix Skill · Owner: Team Ramanathan & Sadhana · a plug-in that makes tool-using AI agents safe

The gap

Voice agents can understand a request and immediately call a refund, cancellation or replacement API. A small missed negation can turn the right conversation into the wrong transaction.

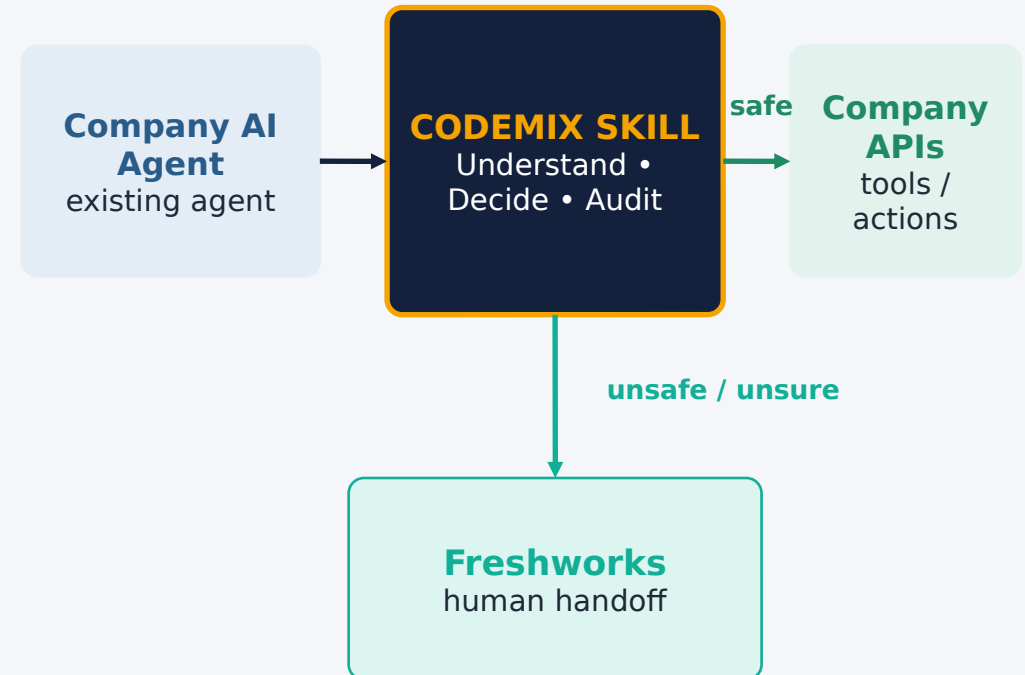
Our product

Codemix Skill sits between the AI agent and its API/tool calls. It validates intent, negation, confidence, sentiment and SOP rules before execution.

Freshworks extension

Unsafe, uncertain or emotionally escalated calls become Freshdesk tickets with transcript, decision context and action details for a human agent.

WHERE IT SITS



The agent proposes an action. Codemix Skill decides whether it runs, needs confirmation, or goes to a human.

2

Problem Statement

In code-mixed voice, one short word can reverse the meaning of an API action

HINDI EXAMPLE

*Caller: "Order cancel **mat** karo, bas address change karna hai."*

Correct intent: do NOT cancel → update address.

TAMIL EXAMPLE

*Caller: "Refund **vendam**, replacement anuppunga."*

Correct intent: do NOT refund → request replacement.

Why this matters Speech-to-text can drop a negation on a noisy line, or an intent model can focus on words like "cancel" or "refund" and ignore the negation. With tool calling, that misunderstanding becomes a real transaction.

RISK

Wrong refund / cancellation

BUSINESS IMPACT

Money, reshipping, trust

NEED

A decision layer before action

3

Why Now

AI agents have write access, so misunderstanding is no longer just a bad answer



AI agents can act

Tool calling and MCP let agents trigger refunds, cancellations, returns and other business actions directly.



Code-mixing is normal

Indian callers switch between languages inside a sentence. The risk is concentrated around small intent-flipping words.



Voice stack is ready

VoBiz provides phone access + DTMF, Sarvam handles Indian speech-to-text, Claude validates intent, and ElevenLabs returns the response.

Our timing The missing layer is not another voice bot. It is a safety and decision layer that controls when an existing AI agent is allowed to call an API.

4

Competitive Landscape

We focus specifically on the gap between language understanding and action execution

Do nothing

- ✓ Zero integration effort.
- ✗ Risky actions can execute on the agent's first interpretation.

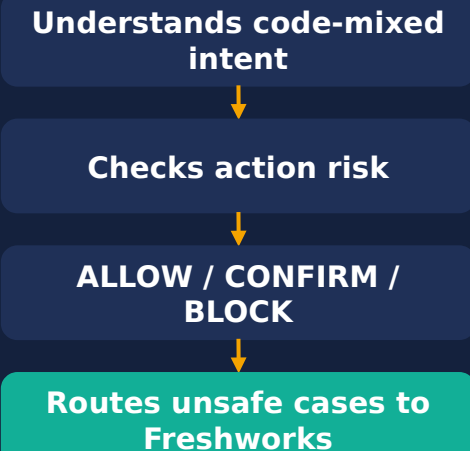
Confirm-everything IVR

- ✓ Safe confirmation.
- ✗ Adds friction to every action, even low-risk requests.

Generic LLM guardrails

- ✓ Useful for policy, toxicity and PII.
- ✗ Not designed as an action gate for code-mixed negation + speech confidence.

Codemix Skill



Friction on every action

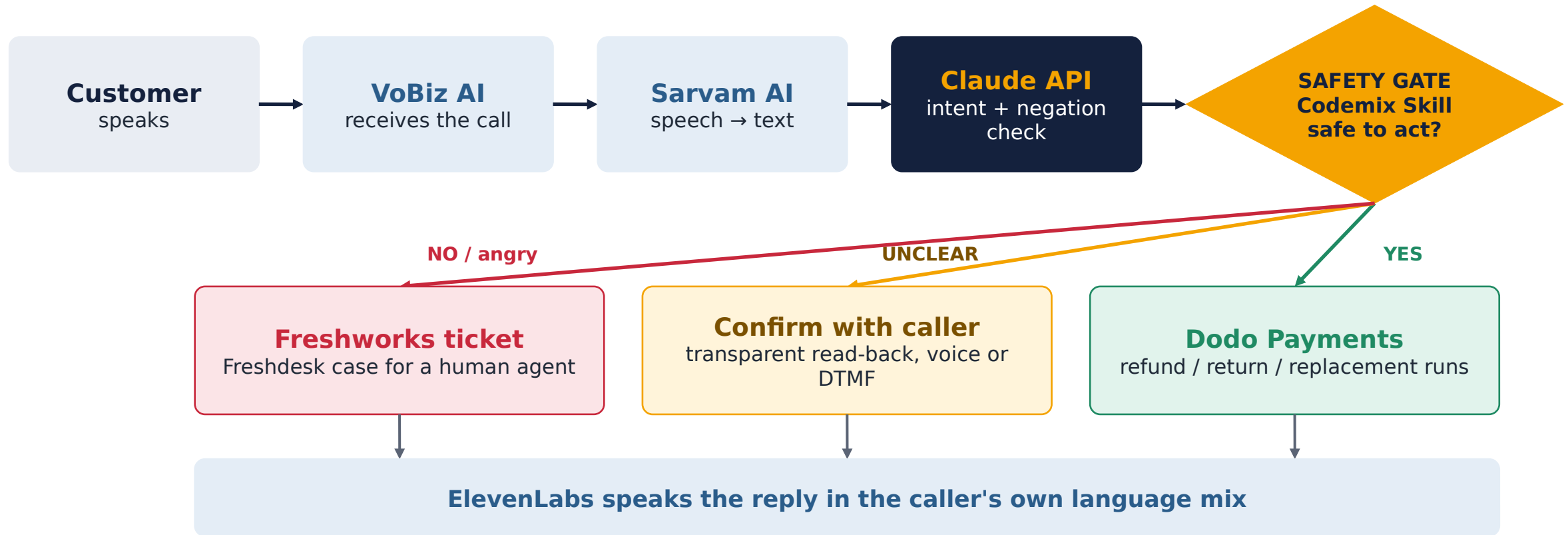
Risk-based: only risky actions get a check

Our edge A policy-aware action gate with human escalation and an auditable record of what the caller said, what the agent decided, and what API was called.

5

Solution Architecture

Codemix Skill is the control point between an existing AI agent and its tools



Our innovation We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

5

Sponsor & Integration Stack

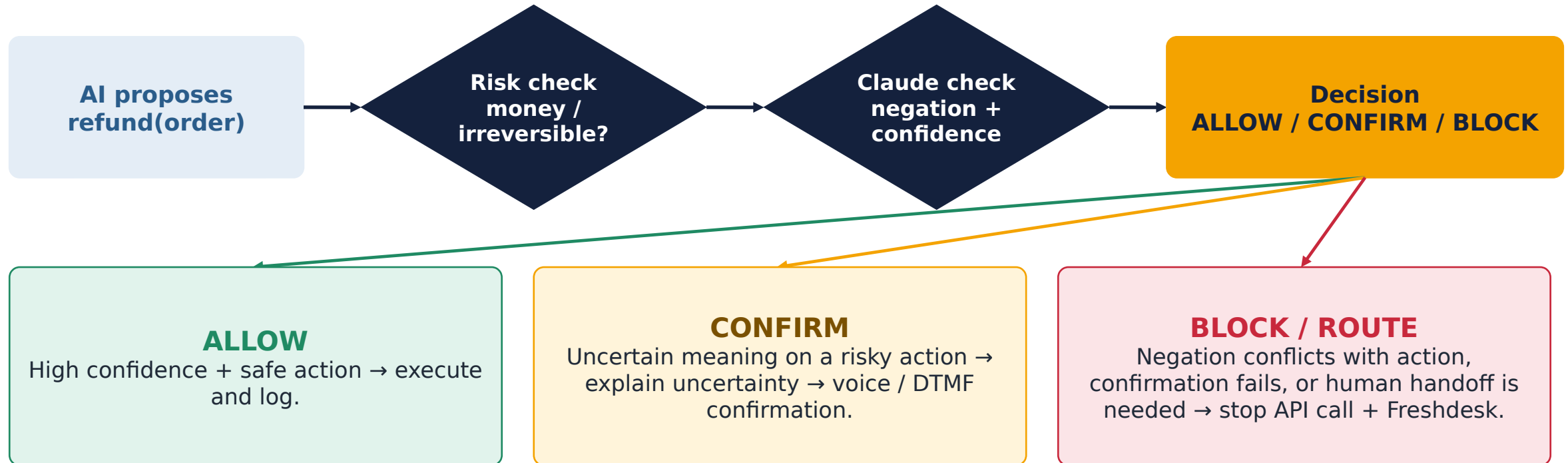
Each sponsor/API has one clear job; Codemix Skill coordinates the handoff and controls action execution

CHANNEL	VoBiz AI	PHONE + DTMF	The number customers call. Keypad (DTMF) gives a confirmation that needs no speech recognition.
LISTEN	Sarvam AI	SPEECH → TEXT	Transcribes code-mixed Hindi + English and Tamil + English, so the negation reaches the intent check.
VALIDATE	Claude API	INTENT + SAFETY	Before any sensitive action: finds the real intent. “Refund vendam” = NO refund, YES replacement.
CONTROL	Codemix Skill	ORCHESTRATION + SAFETY	The safety gate between what the voice agent hears and what it is allowed to do.
ACT	Dodo Payments	ACTION API	Runs refunds, returns and replacements, but only after the safety gate says YES.
ESCALATE	Freshworks / Freshdesk	HUMAN WORKFLOW	If it can't be resolved safely, a Freshdesk ticket hands the case to a human agent.
RESPOND	ElevenLabs	VOICE OUTPUT	Speaks the reply in the caller's own language mix and reads back what it understood.

5

The Safety Decision

The agent never silently guesses on an irreversible action



EXAMPLE

"Refund vendam, replacement anuppunga."

BLOCK refund

Confirm replacement

If needed: Freshdesk ticket

5

The Feature Set

The product combines understanding, decisioning, conversation control and auditability

UNDERSTAND

Mixed-language understanding

Hinglish / Tanglish, including mid-sentence switching.

Intent + negation

Maps what the caller actually wants; catches mat / nahi / vendam / vaddu.

DECIDE

Decision layer

ALLOW, CONFIRM or BLOCK for each API action.

Transparent uncertainty

States what is uncertain and asks before acting.

Sentiment → human

Angry or distressed callers are routed to Freshdesk.

CONVERSE

Low latency

Fast path for routine requests; deeper validation only when sensitive.

DTMF + IVR

Press 1 / 2 confirmation fallback on any phone.

Interrupt handling

Caller can cut in; agent stops speaking and listens.

AUDIT

SOP-based auditing

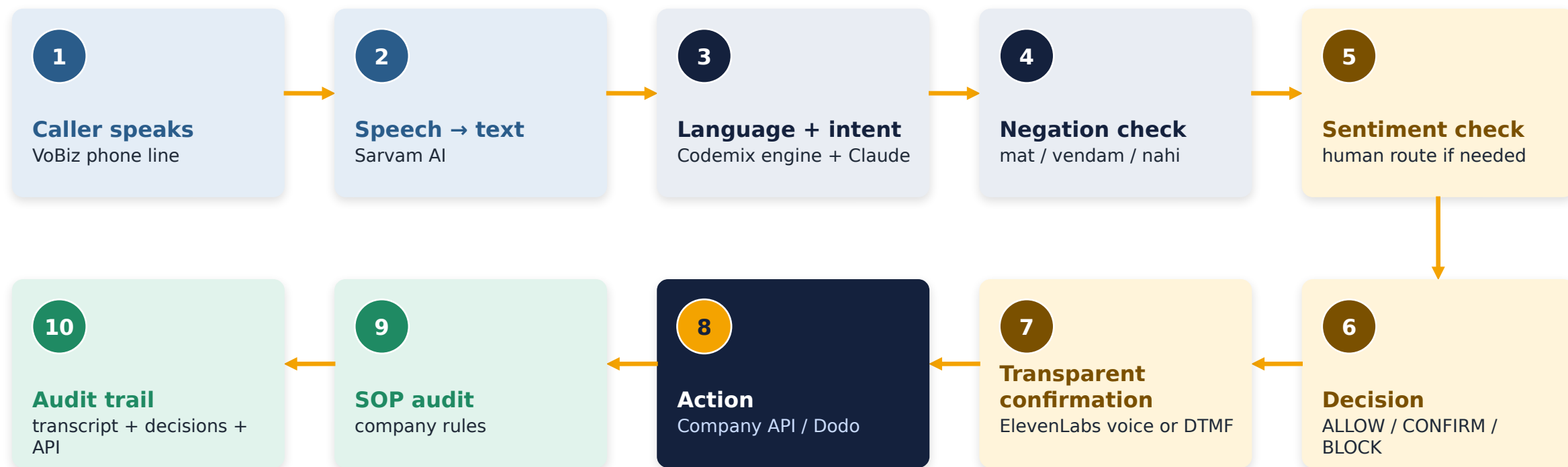
Checks every call against company rules.

Full audit trail

Transcript + decisions + API calls + SOP result.

5 One Call — End to End

Every feature appears only where it adds value

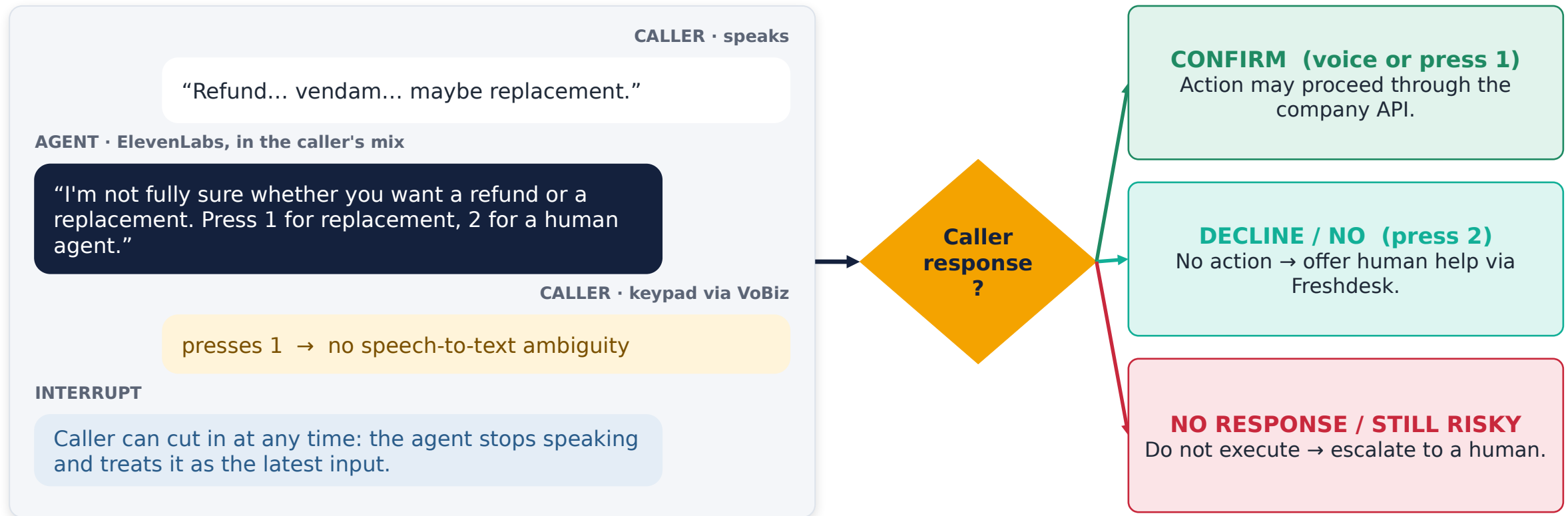


Core principle Most routine calls take the fast path. Confirmation and Freshworks handoff activate only when risk or uncertainty requires them.

5

Confirming Before Acting

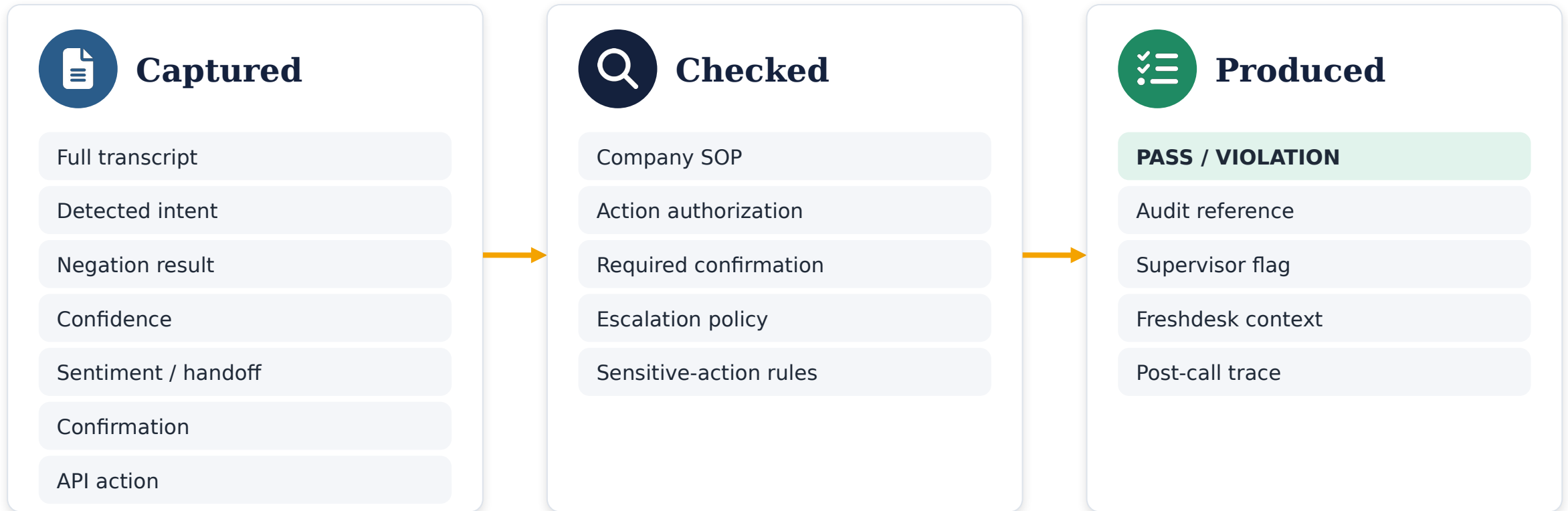
Transparent uncertainty, DTMF / IVR confirmation and interrupt handling in one call



Principle Uncertainty becomes a visible decision point, not a hidden model guess. DTMF gives the safety layer a deterministic input.

5 SOP-Based Auditing & Full Audit Trail

Every sensitive action can be reconstructed after the call



EXAMPLE AUDIT CHAIN



5

Freshworks Integration — Human-in-the-Loop

Freshdesk is where the AI hands over when it should not act alone



When a ticket is created

- Unsafe or unclear action
- Caller does not confirm
- Caller is angry / distressed
- AI should not guess



What we add to Freshdesk

- Code-mixed intent validation
- Action gating (ALLOW / CONFIRM / BLOCK)
- Transparent confirmation
- Structured escalation on top of ticketing, agent workspace and history



What the human agent gets

- English summary + full transcript
- Intent, confidence, safety verdict
- Attempted API action + reason
- A ready-to-handle case, not a blind escalation

Live today `api/create-ticket.js` already creates real Freshdesk tickets. We extend Freshworks at the AI-action boundary; we don't replace Freshdesk or the company's APIs.

5

Freshdesk Ticket Payload

The handoff preserves the evidence a human needs to continue the conversation

Freshdesk ticket · created by Codemix Skill · POST /api/v2/tickets



Conversation context

Customer / call ID

Original transcript

Detected language / code-mix

English summary

Timestamp

Voice / DTMF source



Decision context

Detected intent

Confidence

Negation detected

Safety verdict

Requested API action

Confirmation status



Operational context

Priority / tags

SOP result

Reason for escalation

Attempted API call

Human handoff reason

Audit reference

Judge takeaway Freshdesk is the system where uncertainty becomes a human workflow, with the evidence needed to act quickly.

5

Agentic Integration — The Plugin Control Point

Codemix Skill is middleware, not a replacement AI agent. It intercepts the proposed action before the company API is called

1 EXISTING AI AGENT

Understands the conversation and proposes a tool/API action.

2 CODEMIX SKILL

Receives transcript + proposed action → Claude checks intent and negation; the gate checks risk, SOP and confidence.

intercept

3 DECISION GATE

ALLOW → API call · CONFIRM → customer confirmation · BLOCK → stop action + Freshdesk handoff

4 COMPANY SYSTEMS

Existing APIs execute only after the gate. Freshdesk receives structured human handoffs.

Integration interfaces

MCP / REST adapter around the agent's tool calls. The company keeps its existing agent and business APIs.

Why this matters

One safety + policy layer can sit in front of multiple action APIs without rebuilding the voice agent.

Core positioning Plugin between the AI agent and its action tools.

5 MVP Scope & What Is Already Built

Separate what is demonstrated today from the intended MVP and future extension

DEMONSTRATED TODAY

- Round-1 code-mix tagging engine
- Intent scoring
- MCP server
- Live Freshdesk ticket API
- Vercel demo
- Freshdesk ticket creation flow

MVP SCOPE

- VoBiz + Sarvam + ElevenLabs voice path
- Claude safety gate
- Dodo action path
- Freshdesk escalation
- Hindi + Tamil + English
- Transcript + audit trail

NEXT PHASE

- Sentiment handoff tuning
- Interrupt handling tuning
- SOP rule editor + audit reports
- More Indian languages
- Freshworks Marketplace app
- Plug-in via MCP / REST

Credibility rule Any latency, accuracy or adoption number that has not been measured in the demo is presented as a target, not a result.

6

Target Customers

Start where AI agents already have permission to take customer actions



Primary customers

D2C, e-commerce and fintech brands using Freshdesk and deploying AI voice agents for Indian customers.



Secondary customers

BPOs and support outsourcers that need one safety / policy layer across multiple client accounts.



Protected actions

Refunds • returns • replacements • cancellations • other company APIs exposed to the AI agent.

WHO WE START WITH

Freshdesk customers

...running AI voice agents

...with Indian, code-mixed callers

...whose agent can refund / return / replace

The narrower the funnel, the higher the cost of a wrong action.

7 Value Proposition

Why a customer would choose this over doing nothing or a generic guardrail

Plug it in once: the existing AI agent handles mixed-language customers, and nothing irreversible happens on a misheard word.

✗ Without Codemix Skill

- Agent acts on its first interpretation
- A dropped “mat” or “vendam” becomes a wrong refund or cancellation
- Blind escalation: humans start from zero
- No record of what the caller said vs. what the agent did

✓ With Codemix Skill

- Validates intent and negation before any sensitive action
- Confirms when uncertain (voice or DTMF)
- Routes people to Freshworks with full context
- Records the full audit trail: transcript, decision, API call, SOP result

8

Pricing & Packaging

Proposed pricing, validated in the pilot. Only guarded (sensitive) actions are metered

Monitor mode

Free

for every Freshdesk customer

- Flags risky actions, never blocks
- Weekly wrong-action report
- Shows the value before paying

Codemix Skill add-on

\$5

per 1,000 guarded actions

- Full safety gate: ALLOW / CONFIRM / BLOCK
- DTMF confirmation + interrupts
- Freshdesk escalation + audit trail
- First 30 days free

Enterprise

Custom

annual contract

- Custom SOP rule packs
- Transcript retention + data residency in India
- SLA

Price tracks the risk removed: routine requests are free; only refunds, returns, replacements and cancellations count.

9 Go-to-Market Plan

A simple commercial path: pilot → measurable proof → Freshworks distribution



Why Freshworks matters Freshdesk converts an unsafe/uncertain AI decision into an actionable, context-rich human support case, which is what makes the add-on sellable.

10

Costs & Resources Needed

Proposed budget for a 3-month MVP



3 engineers

voice path, safety gate, Freshworks app



1 linguist / data annotator

Hindi + Tamil negation set



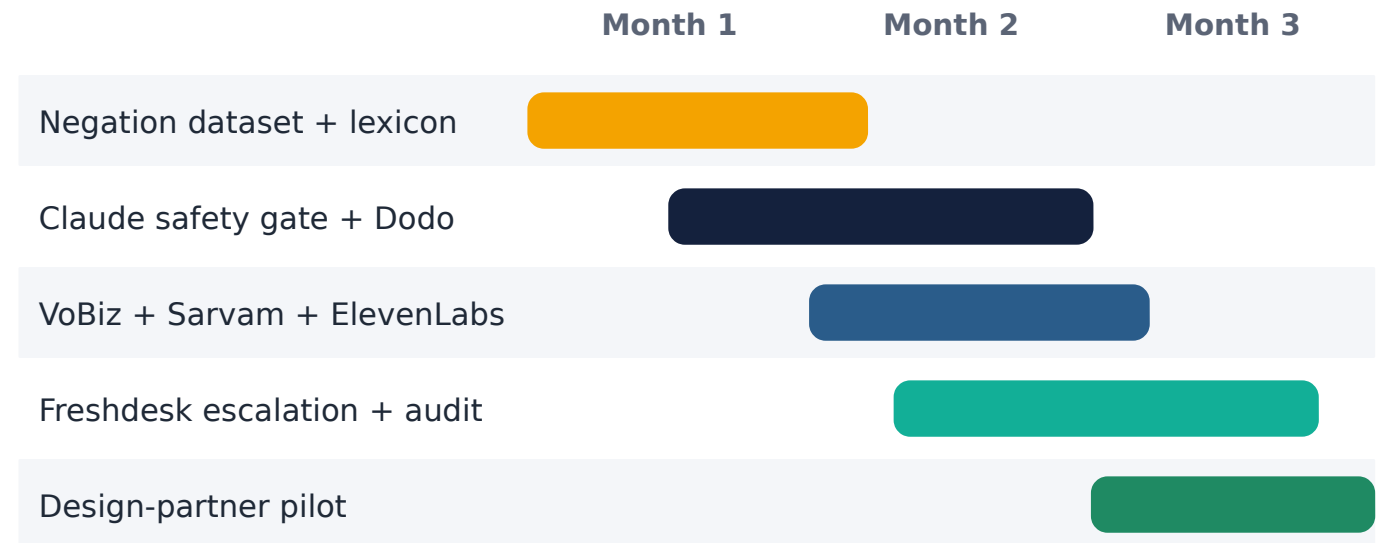
Part-time PM

pilots + pricing

~\$90K

proposed MVP budget: salaries, Sarvam / Claude / ElevenLabs / VoBiz usage, negation data labelling

3-month MVP timeline



Optional: revenue estimate (assumption-based)

100 paying customers × 20,000 guarded actions / month × \$5 per 1,000
 ≈ **\$120K ARR in Year 1**. Every input is an assumption to test in the pilot.

11

Expected Benefits & Success Metrics

Targets to validate in the pilot, not measured results

< 0.5%

Wrong actions

on negated Hindi + Tamil requests

< 900 ms

p95 reply time

on the sensitive-action path

≤ 15%

Confirmation rate

friction budget

100%

Audit coverage

transcript + decision + API + SOP

–30%

Reversal tickets

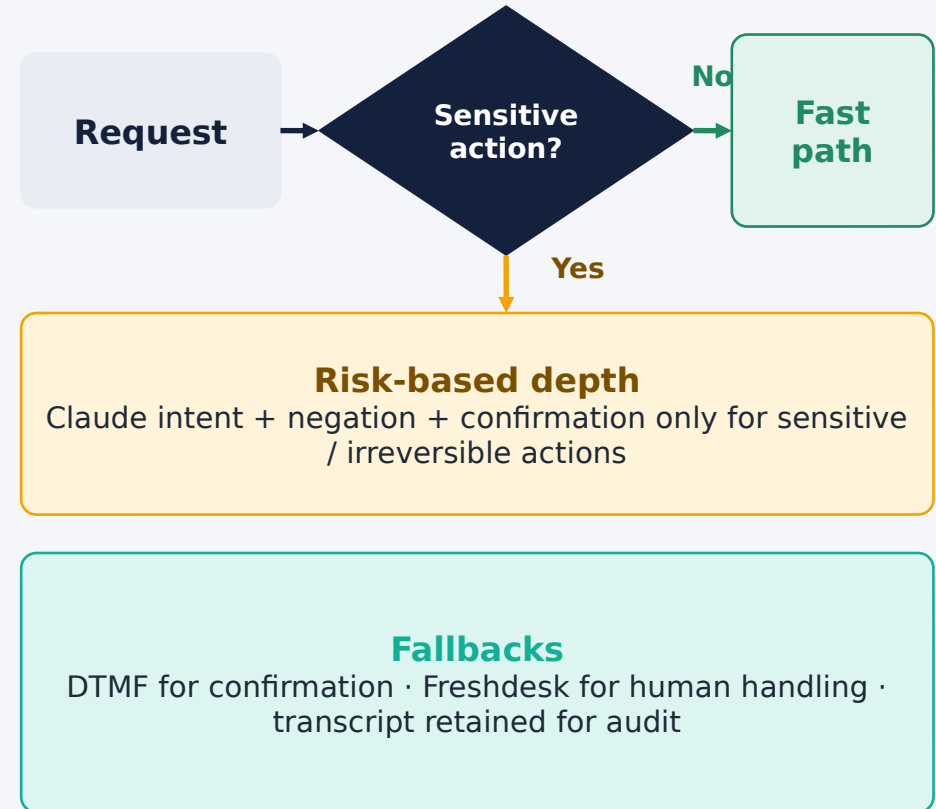
at pilot customers

15%

Adoption

of eligible customers in 2 quarters

HOW LOW LATENCY IS KEPT

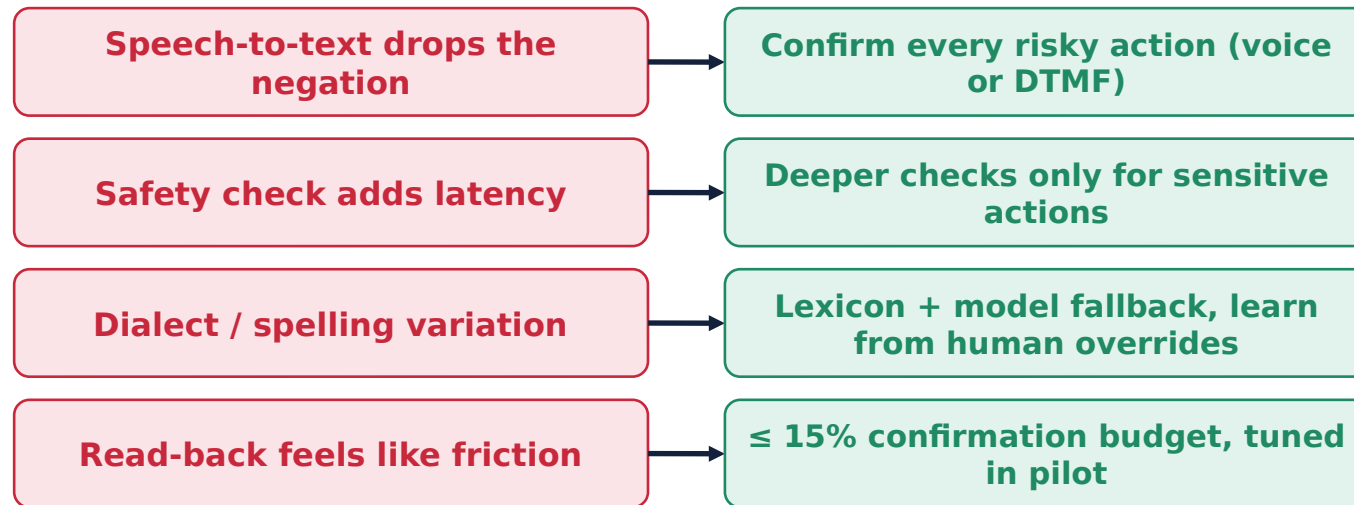


12

Risks & Assumptions

What could go wrong, how we reduce it, and what we're assuming

KEY RISKS → MITIGATION



ASSUMPTIONS

- Pilot brands expose refund / return APIs to their AI agent
- Callers accept a short confirmation on risky actions
- Sarvam accuracy on phone audio is good enough with the confirmation fallback

OUR ASK

Freshdesk sandbox + Marketplace listing path for a 3-brand pilot.

Use the pilot to measure wrong-action reduction, confirmation rate, latency and adoption.

Decision principle

Do not silently guess on a sensitive action; expose uncertainty, confirm or route to a human.



Final Pitch — Why This Is a Freshworks Extension

A concise closing story for the judges

THE PROBLEM

Code-mixed voice + negation can cause an AI agent to misunderstand an action.

THE PRODUCT

An intent-validation safety gate between what the voice agent hears and what it is allowed to do:
Understand → Decide → Act → Audit.

THE FRESHWORKS VALUE

When the AI should not act, Freshdesk turns uncertainty into a structured human workflow with transcript + decision context.

THE DEMO



“Don't cancel” should mean don't cancel — and Freshworks should know exactly what happened when the AI chooses not to act.

Codemix Skill • Understand → Decide → Act → Audit • Thank you